

AI & Data Analytics Assessment

Global incident intelligence · Risk prioritization · Response operations · Explainable machine learning

Prepared by	Shashank		
Dataset	2,000 physical security incidents across 8 global sites		
Coverage	2025-01-01 to 2026-06-30		
Deliverables	Interactive Streamlit dashboard, reproducible Python pipeline, model, and report		

Purpose

Convert a supplied assessment dataset into defensible operational insight and demonstrate how a governed AI triage model could support—not replace—physical security decision-making.

1. Executive summary

Incidents	High / critical	Critical	Avg response	Confirmed
2,000	776 (38.8%)	165	11.6 min	55.6%

- Austin has the highest raw incident volume (269), but volume is not a population-adjusted incident rate.
- Dublin records the most critical incidents (30).
- Dublin ranks first on the transparent comparative risk index (86.8/100).
- Dublin has the slowest average response (11.88 minutes).
- Lost Badge is the most frequent incident type (207 records).
- Suspicious Package has the highest observed high/critical share (100.0%).
- Night shift has the slowest average response (11.93 minutes).
- 2026-01 has the highest monthly incident count (129).

Leadership interpretation

Dublin warrants the first management review because it combines the highest critical count, the highest comparative risk score, and the slowest average response. Austin has the highest volume, but without site-exposure denominators it should not automatically be labelled the least safe location.

Recommended action

Pilot an AI-assisted intake queue with human approval, establish severity-specific response SLAs, investigate site-level drivers, enrich the incident schema, and prospectively validate performance before operational deployment.

2. Data quality and preprocessing

Check	Result	Treatment
Schema	12/12 expected fields	Standardized names; original CSV preserved
Completeness	0 blank cells	No imputation required
Uniqueness	0 duplicate IDs / rows	Incident ID retained only for traceability
Validity	0 invalid dates or response times	Typed date and numeric response fields
Categories	0 unexpected values	Explicit category validation
Badge Access N/A	90 records	Retained as a distinct category; meaning is ambiguous
Quality score	100.0/100	Transparent issue-count score, not a universal standard

Quality caution

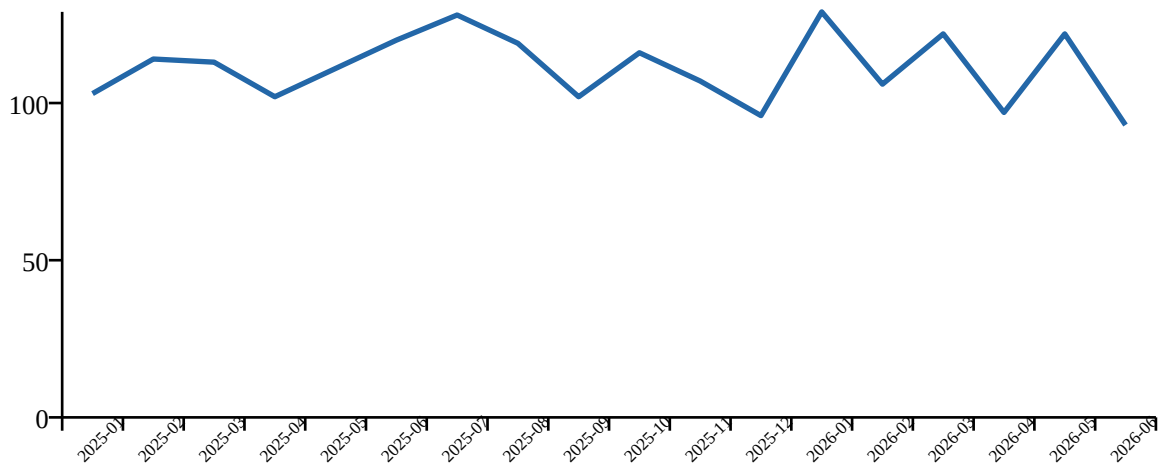
The file is unusually complete, consistent, and balanced for operational incident data. It may be curated or synthetic for assessment purposes. The analysis therefore emphasizes method, transparency, and limitations rather than treating small observed differences as causal evidence.

Engineered fields

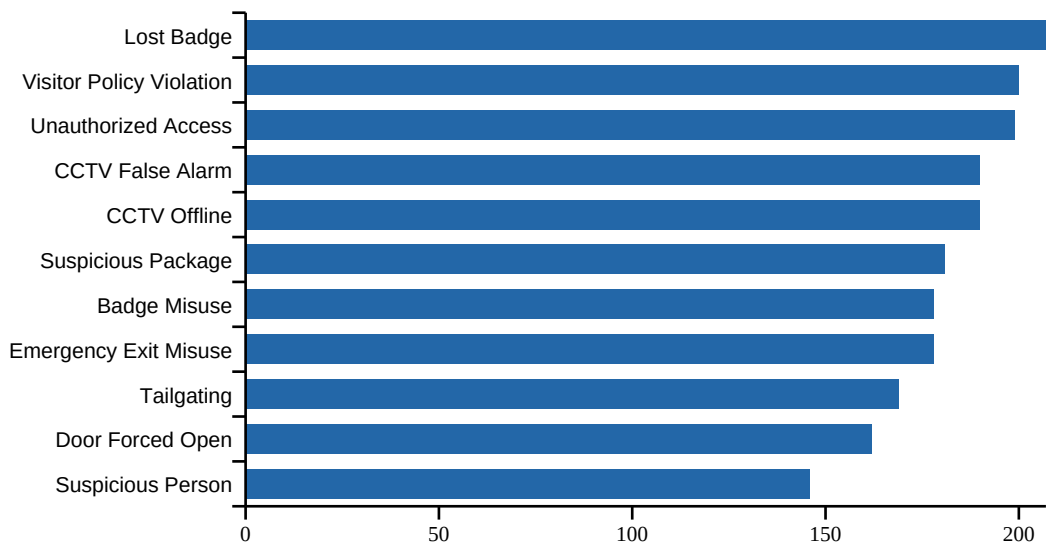
- Calendar features: year, month, year-month, weekday, and weekend indicator.
- Operational flags: high/critical target, confirmed, under investigation, false alarm, and response above 15 minutes.
- Severity weights used only in descriptive prioritization—not as model inputs.

3. Exploratory findings

Monthly incident volume



Incident volume by type

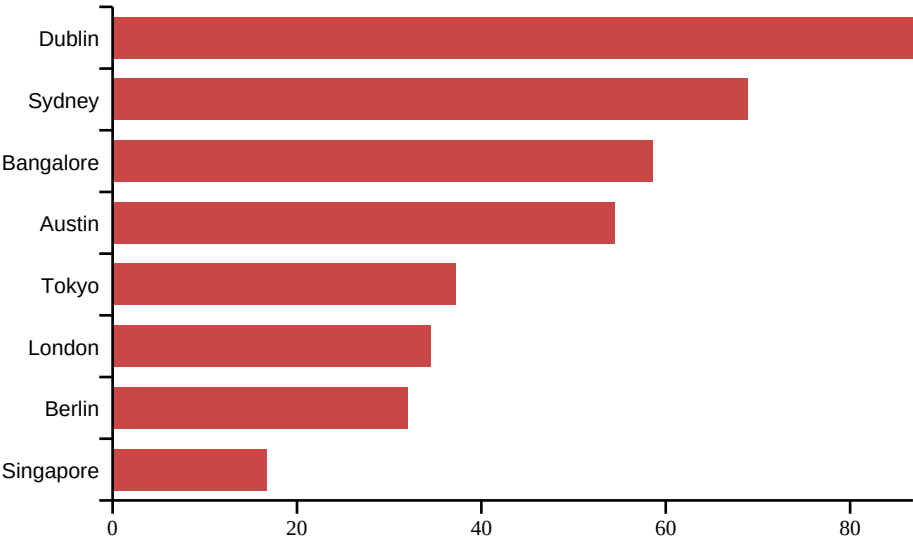


Interpretation

Lost Badge is the largest incident category. Suspicious Package and Door Forced Open are entirely labelled high/critical in this sample, while several categories never receive those labels. This strong label structure is operationally useful for rule-based escalation but also explains much of the model's predictive power.

4. Site risk and response operations

Transparent comparative site risk score



Rank	Site	Incidents	Critical	High-risk %	Avg response	P90 response	Score
1	Dublin	241	30	44.0%	11.9	18.0	86.8
2	Sydney	252	18	42.9%	11.8	18.0	69.0
3	Bangalore	237	21	38.8%	11.7	18.0	58.6
4	Austin	269	20	38.3%	11.6	18.0	54.4
5	Tokyo	248	19	33.9%	11.7	18.0	37.2
6	London	260	16	38.1%	11.5	18.0	34.6
7	Berlin	259	24	39.0%	11.2	17.0	32.0
8	Singapore	234	17	35.5%	11.3	17.0	16.8

Risk-index formula

35% high/critical share + 20% critical count + 20% 90th-percentile response + 15% under-investigation share + 10% positive recent trend. Each component is min-max normalized across the eight sites. It is a comparative prioritization index—not a calibrated probability of harm.

Response finding

Overall response averages 11.59 minutes; median is 12 and the 90th percentile is 18. A 15-minute threshold is shown only as an analytical scenario because no official SLA was supplied.

5. AI/ML approach

Use case

At incident intake, estimate the probability that the case will be classified High or Critical and prioritize cases above an approved decision threshold for human review.

Design choice	Implementation
Target	1 = High/Critical; 0 = Low/Medium
Inputs	Site, incident type, shift, badge, CCTV, visitor, month, weekday, weekend
Leakage exclusions	Severity, outcome, response time, and Incident ID
Validation	Earliest 60% train; next 20% validation; latest 20% untouched test
Threshold	0.55; maximize validation recall subject to precision ≥ 0.70
Selected model	Random Forest
Human role	Model recommends queue priority; security professional owns escalation

Model	Recall	Precision	F1	ROC-AUC	PR-AUC	Brier
Random Forest	0.818	0.704	0.756	0.917	0.879	0.121
Logistic Regression	0.811	0.723	0.764	0.920	0.880	0.112
Gradient Boosting	0.804	0.708	0.753	0.913	0.869	0.105
Dummy Baseline	0.000	0.000	0.000	0.500	0.370	0.234

Latest-period test result

The selected model identifies 121 of 148 high-risk incidents (recall 81.8%) while 51 lower-risk cases are also escalated. Precision is 70.3%; PR-AUC is 0.877; ROC-AUC is 0.917.

6. Validation, governance, and limitations

Why these metrics?

- Recall measures how many truly high-risk cases are caught; a missed critical incident is costly.
- Precision measures alert burden and guards against alert fatigue.
- F1 balances recall and precision at the operating threshold.
- PR-AUC evaluates ranking quality for the positive operational class.
- ROC-AUC measures overall separation; Brier score evaluates probability accuracy/calibration.
- The confusion matrix translates model scores into operational case volumes.

Required production controls

- Prospective shadow-mode validation on future incidents.
- Monthly monitoring of recall, precision, calibration, alert volume, and drift.
- Site-level performance checks to identify uneven error rates.
- Versioned data, model, threshold, and approvals with an audit trail.
- Human override, escalation documentation, and periodic threshold review.

Limitations

- No site population, employee count, visitor volume, floor area, or operating-hour exposure is supplied; raw site counts are not true incident rates.
- No precise event timestamp, dispatch timestamp, resolution duration, loss amount, narrative, or guard staffing data is available.
- The unusually complete and balanced sample may be curated or synthetic; small differences should not be treated as causal evidence.
- The risk index is a transparent prioritization aid, not a probability of harm or a replacement for security judgment.

Most important modelling caveat

Incident type strongly encodes the supplied severity labels. Performance is therefore an assessment-sample proof of concept, not evidence of production generalization. Validate prospectively on future operational data before deployment.

7. Actionable roadmap

Horizon	Action	Success measure
0-30 days	Define severity-specific SLAs; review Dublin critical cases; standardize closure reasons.	Approved SLA and site action plan
30-60 days	Run AI triage in shadow mode; compare model queue with analyst decisions.	Recall, precision, alert load and overrides
60-90 days	Add timestamps, zones, staffing, footfall, impact, narratives and exposure denominators.	Improved data completeness and normalized rates
90+ days	Controlled pilot with human approval, drift monitoring and quarterly governance review.	Faster high-risk response without alert fatigue

Final recommendation

Treat the model as a prioritization layer within an accountable operating process. The strongest near-term value comes from combining transparent analytics, consistent response targets, richer incident capture, and human-reviewed risk scoring—not from autonomous security decisions.

Supporting artefacts

- Interactive Streamlit dashboard with filters and intake prediction.
- Reproducible cleaning, analysis, model training, and report scripts.
- Cleaned dataset, quality report, site summaries, model artefacts, and test predictions.
- README, notebook walkthrough, and reproduction instructions.